

E. coli Transformation

A protocol for transforming competent E. coli cells with plasmid DNA, including heat shock and plating on selective media.

Equipment

- pipette
- ice bucket (green container with ice)
- thermal cycler (used for heat shock at 42°C)

Reagents

- plasmid DNA
- competent E. coli cells
- LB agar plate (containing appropriate selective antibiotic)

Procedure

1. Add 5 μ L of plasmid DNA to the competent E. coli cells.
2. Mix gently by flicking the tube 4-5 times.
3. Incubate the DNA-cell mixture on ice for 5 seconds.
4. Rapidly transfer the tube to a pre-heated 42°C water bath for exactly 5 seconds.
5. Immediately return the tube to ice for 5 seconds.
6. Spread 50 μ L of culture onto an LB agar plate containing the appropriate selective antibiotic.
7. Invert the plates.